

10/9/2026: Contrapositive and contradiction

CSCI 246: Discrete Structures

Textbook reference: Sec 20, Scheinerman

Today's Agenda

- Mini lecture (≈ 25 mins)
- Group exercises (≈ 25 mins)

Overview: Three ways to prove implications

Contrapositive: Details

Contradiction: Details

Applications

There are three ways to prove $P \implies Q$.

Poll. Suppose P and Q are propositions. How can we prove $P \implies Q$?

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1. **By direct proof.** This is what we've been doing all along. See Proof Template 1, Scheinerman pp. 17.
2. **By contraposition.** We can prove $\neg Q \implies \neg P$.
3. **By contradiction.** We can assume $\neg(P \implies Q)$ and find a contradiction. (Note: this is the same as assuming $P \wedge \neg Q$ and finding a contradiction.)

One claim, three proofs

Claim. Let n be an integer. If $5n + 3$ is even, then n is odd.

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Direct proof: Show $P \implies Q$. **Contraposition:** Show $\neg P \implies \neg Q$. **Contradiction:** Show $P \wedge \neg Q$ is impossible.

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Direct proof.

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Direct proof. Assume $5n + 3$ is even, so $5n + 3 = 2k$ for some integer k . Then $n = 2k - 4n - 3 = 2(k - 2n - 2) + 1$, which is odd because $k - 2n - 2$ is an integer. ■

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By contraposition. The contrapositive is: if n is even, then $5n + 3$ is odd. Let n be even, so $n = 2k$. Then $5n + 3 = 10k + 3 = 2(5k + 1) + 1$, which is odd because $5k + 1$ is an integer. ■

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By contradiction. Suppose, for contradiction, that $5n + 3$ is even but n is even. Since n is even, we write $n = 2k$. But then $5n + 3 = 10k + 3 = 2(5k + 1) + 1$, which is odd. So $5n + 3$ is both even and odd, which is impossible. ■

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implications

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Contradiction: Details

Applications

Contraposition: Why does it work?

Poll. How to show that the contrapositive technique works?

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Solution. Recall that the truth table for the $\implies$ operator is given by

P	Q	$P \implies Q$
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T	F	F
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Now what?

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Now we apply the $\implies$ operator to inputs transformed by the "not" operator (also written $\neg$).

P	Q	$\neg Q$	$\neg P$	$\overbrace{\neg Q \implies \neg P}^{\text{If not Q, then not P}}$
T	T	F	F	T
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F	T	F	T	T
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Solution. Recall that the truth table for the $\Rightarrow$ operator is given by

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The two tables have the same final column!

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Hence $P \implies Q$ is equivalent to $\neg Q \implies \neg P$.

Contraposition: When should we use it?

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Claim. Let a, b be integers. If ab is even, then a is even or b is even.

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Proof. Write $a = 2k + 1$ and $b = 2\ell + 1$. Then

$$\begin{aligned} ab &= (2k + 1)(2\ell + 1) = 4k\ell + 2k + 2\ell + 1 \\ &= 2(2k\ell + k + \ell) + 1, \end{aligned}$$

which has the form $2m+1$, i.e. odd. So the original claim holds.

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Conclusion. We can prove $P \implies Q$ by assuming $P \wedge \neg Q$ and finding a contradiction.

Contradiction: Another example

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Claim. If n^2 is even, then n is even.

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Proof by contradiction. We assume $P \wedge \neg Q$, where

- P : n^2 is even
- $\neg Q$: n is odd.

and derive a contradiction.

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Now:

- If n is odd, write $n = 2k + 1$.
- Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd. $\Rightarrow \Leftarrow$

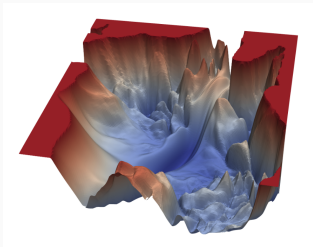
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Applications

Machine learning: Why gradient descent stops

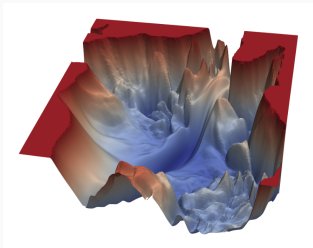


A nonconvex loss surface for a neural network: many local minima and saddle points where gradient descent can stop.

Source: H. Li, Z. Xu, G. Taylor, C. Studer, & T. Goldstein, "Visualizing the Loss Landscape of Neural Nets," *NeurIPS* 2018.

Fact. If x is a local minimum of a differentiable function f , then $\nabla f(x) = 0$.

Machine learning: Why gradient descent stops



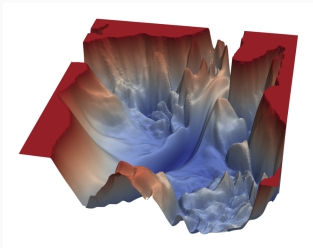
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Fact. If x is a local minimum of a differentiable function f , then $\nabla f(x) = 0$.

Contrapositive. If $\nabla f(x) \neq 0$, then x is not a local minimum, so you can still improve by stepping downhill.

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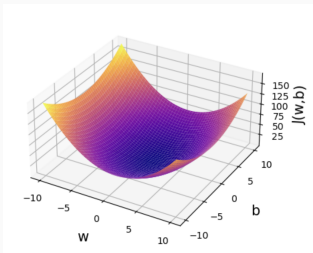
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Remark. The second form is the version gradient descent actually uses: keep stepping while the gradient is nonzero.

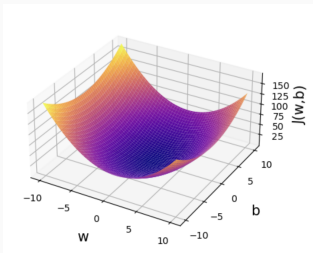
Machine learning: Convex losses have no bad local minima



A convex loss surface $J(w, b)$ (e.g., squared error for linear regression).

Claim. If a function f is convex, every local minimum of f is a global minimum.

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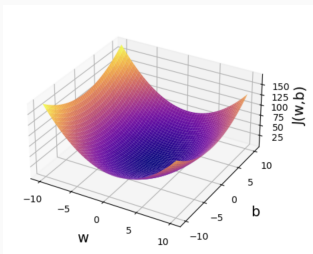


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Claim. If a function f is convex, every local minimum of f is a global minimum.

Proof sketch. By contradiction. Suppose x^* is a local minimum but not a global one. Then some y has $f(y) < f(x^*)$. Convexity says f lies below the chord on the segment from x to y , so points on that segment arbitrarily close to x^* have values below $f(x)$. That contradicts x being a local minimum. ■

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Who cares. Linear regression and logistic regression have convex losses. This short contradiction argument guarantees that gradient descent can't get stuck at a bad solution on those models.

Group exercises

Ashley, Isabelle: 4
Bailey, Tanna: 9
Blaisdell, Olyver: 3
Brown, Ashton: 3
Brown, Jonathan: 11
Cardenas, Dominic: 12
Cutler, George: 8
Fabik, Roland: 7
Fellows, Audrey: 6
Fitzgerald, Ryan: 1
Fuller, Ryan: 12
Griffen, Clara: 9
Grutsch, Tanner: 7

Hager, Nathan: 8
Horner, Jayden: 13
Johnson, Jaret: 9
Kintzel, Samantha: 5
Lameres, Alexis: 2
Lee, Holden: 1
Mayala, Reece: 12
McLean, Connor: 3
Mcbride, Daniel: 2
Miller, Maverick: 4
Moss, Emily: 8
Mousseau, Matthew: 7

Neuman, Evan: 6
Putnam, John: 5
Ramos, Vydel: 2
Rathbun, Jedidiah: 10
Roberts, Matthew: 13
Roller, Nathan: 11
Ryan, Otis: 4
Shevtsov, Timothy: 1
Sinclair, Aidan: 5
Stensland, Emma: 10
Thiessen, Logan: 6
Thompson, Jaiden: 10
Vath, Ethan: 11

Group exercises

1. State the contrapositive of each of the following statements.
 - a. If it is raining, then the ground is wet.
 - b. If the battery is fully charged, the car will start.
 - c. If n^2 is even, then n is even.
2. Prove by contraposition that if n^2 is even, then n is even.
3. What is the contrapositive of a contrapositive of an if-then statement?
4. Prove by contradiction that consecutive numbers cannot be both even.

Solution to group exercise #1

Problem. State the contrapositive of each of the following statements.

- a. If it is raining, then the ground is wet.
- b. If the battery is fully charged, the car will start.
- c. If n^2 is even, then n is even.

Solution.

- a. If the ground is not wet, then it is not raining.
- b. If the car does not start, then the battery is not fully charged.
- c. If n is odd, then n^2 is odd.

Solution to group exercise #2

Problem. Prove by contraposition that if n^2 is even, then n is even.

Solution.

Contrapositive. If n is odd, then n^2 is odd.

Proof of contrapositive. Write $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is of the form $2m + 1$, i.e. odd. Thus the contrapositive is true, so the original claim holds.

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which is of the form $2m + 1$, i.e. odd. Thus the contrapositive is true, so the original claim holds.

Remark. The direct route (“If n^2 is even, then n is even.”) is awkward to attempt directly; the contrapositive is immediate by simple algebra.

Solution to group exercise #3

Problem. What is the contrapositive of a contrapositive of an if-then statement?

Solution.

Original statement. If P then Q .

Contrapositive. If $\neg Q$, then $\neg P$.

Contrapositive of contrapositive. If $\neg\neg P$, then $\neg\neg Q$. In other words, we recover the original statement.

Solution to group exercise #4

Problem. Prove by contradiction that consecutive numbers cannot be both even.

Solution. Suppose by way of contradiction that x and $x + 1$ are both even. Then by definition of even, $x = 2a$ and $x + 1 = 2b$ for some integers a and b . By substitution, we have

$$2a + 1 = 2b \implies 2b - 2a = 1 \implies b - a = \frac{1}{2}.$$

But it's impossible for the difference of two integers to be $\frac{1}{2}$. $\Rightarrow \Leftarrow$.